



Approach To Primary And Secondary Amenorrhea

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Objectives :

- Differentiate between the primary and secondary amenorrhea.
- Understanding the etiology of both primary and secondary amenorrhea.
- How to approach for the management of primary and secondary.

Amenorrhoea is defined as the absence of menstruation for more than 6 months in the absence of pregnancy in a woman of fertile age.

oligomenorrhoea is defined as irregular periods at intervals of more than 35 days, with only 4–9 periods a year.

primary amenorrhoea : no menstruation till 16 years of age.

Secondary amenorrhoea is absence of menstruation for more than six months in a normal female of reproductive age that is not due to pregnancy, lactation or the menopause.

- From a clinical point of view it is probably best to classify the a etiologies of **primary amenorrhoea** based on the presence or absence of secondary sexual characteristics.

If the Secondary sexual characteristics are normal the possible causes are:

- Imperforate hymen
- Transverse vaginal septum
- Absent vagina and functioning uterus
- Absent vagina and non-functioning uterus
- XY female: androgen insensitivity
- Resistant ovary syndrome
- Constitutional delay

If the **Secondary sexual characteristics are absent**, so the classification will be according to the patient height as below:

If the patient had **Normal stature**

1-Hypogonadotrophic hypogonadism

- Congenital(Isolated gonodotrophin-releasing hormone deficiency, Olfactogenital syndrome)
- Acquired (Weight loss/anorexia,Excessive exercise,Hyperprolactinaemia)

2-Hypergonadotrophic hypogonadism

- Gonadal agenesis as XX agenesis or XY agenesis,
- Gonadal dysgenesis as Turner mosaic,Other X deletions or mosaics, XY enzymatic failure
- Ovarian failure
- Galactosaemia

If the patient had **Short stature**

1-Hypogonadotrophic hypogonadism

- Congenital as Hydrocephalus
- Acquired as Trauma, Empty sella syndrome, Tumours.

2-Hypergonadotrophic hypogonadism

- Turner syndrome and Other X deletions or mosaics

Heterosexual development

- Congenital adrenal hyperplasia
- Androgen-secreting tumour
- 5 α -Reductase deficiency
- Partial androgen receptor deficiency
- True hermaphrodite
- Absent Müllerian inhibitor

Imperforate hymen

- It may present in early childhood when the infant presents with a bulging hymen behind which is a **mucocele**, or at puberty when there is cyclical pain due to collection of blood in the vagina (**hematocolpus**), may be associated difficulty with micturition and defecation.
- the management will be by hymenectomy .



Transverse vaginal septum

- The vagina fails to cannulate, the upper and lower parts of the vagina are separate.
- cyclical abdominal pain due to the haematocolpos
- There will be abdominal mass (hematometra or hematocolpus), The hymenal remnants are often seen separately.
- Transverse vaginal septum may occur at three levels, known as a lower, middle or upper third septum.
- The management is very different from imperforate hymen and very careful assessment must be made before embarking on any management strategy.

Absent vagina and the functioning uterus

- The uterine body has developed normally but there is failure of development of the cervix and upper vagina. cyclical abdominal pain, but there is no pelvic mass to be found because there is no vagina to be distended.
- retrograde menstruation occurs leading to the development of endometriosis and in some patients pelvic adhesions.

Absent vagina and a non-functioning uterus(Mayer–Rokitansky–Küster–Hauser (MRKH) syndrome)

- 2nd most common cause of primary amenorrhoea, Secondary sexual characteristics are normal as would be expected as ovarian function is unaffected.
- Examination of the genital area discloses normal female external genitalia but a blind-ending vaginal dimple which is usually not more than 1.5cm in depth.

XY female mention in the lecture of DSD.

Resistant ovary syndrome

- extremely rare cause of primary amenorrhoea.
- There are elevated levels of gonadotrophin in the presence of apparently normal ovarian tissue; patients do have some development of secondary sexual characteristics, but never produce adequate amounts of oestrogen to result in menstruation.
- It is believed that these women have an absence or malfunction of FSH receptors in the ovarian follicles, and are unable to respond properly to FSH.

Constitutional delay

- There is no anatomical anomaly and endocrine investigations are all normal.
- This is because of immature pulsatile release of GnRH.
- These young women will eventually menstruate spontaneously as the maturation process proceeds.

Absent secondary sexual characteristics (normal height)

Isolated GnRH deficiency (olfactogenital syndrome, Kallman's syndrome)

- the hypothalamus lacks the ability to produce GnRH and there is therefore a hypogonadotrophic state. maldevelopment of neurones in the arcuate nucleus of the Hypothalamus, also it can be associated with anosmia.
- The pituitary gland is normal and stimulation with exogenous GnRH leads to normal release of gonadotrophins.

Weight loss/anorexia:

- young girls may suffer from anorexia nervosa in the prepubertal state; this leads to failure of activation of the gene which initiates GnRH release in the hypothalamus, and therefore a persistent hypogonadotrophic state exists.
- The growth spurt is not usually influenced by this, but secondary sexual characteristics are absent.

Excessive exercise

- excessive exercise in pubertal children leads to decreased body fat content, without necessarily affecting body mass, like ballet dancers, athletes and gymnasts.

Hyperprolactinaemia

- There may be a recognizable prolactinoma in the pituitary.

Gonadal agenesis

- There is complete failure of development of the gonad. These girls may be either 46XX or 46XY.
- Genotype does not affect their phenotype, all of them being female.
- Height is normal as the growth spurt occurs at the normal time.
- In those girls who are 46XY, the failure of production of androgen or oestrogen means that their long bones do not undergo epiphyseal closure at the normal time and therefore final height may be excessive.

Ovarian failure

- either chemotherapy or radiotherapy for childhood malignancy.

Galactosaemia

- This inborn error of galactose metabolism is due to deficiency of galactose-1-phosphate uridylyltransferase.
- Those patients have an acute toxic syndrome that causes ovarian cellular destruction thought to be due to the accumulation of galactose metabolites, which may induce programmed cell death (apoptosis).

Gonadal dysgenesis

- The gonad is described as dysgenetic if it is abnormal in its formation.
- The commonest is Turner's syndrome,
- In XY individuals there may be a dysgenetic gonad associated with enzymatic failure, no production of testosterone The external phenotype is therefore female with a short vagina.

Absent secondary sexual characteristics (short stature)

Congenital infection

- As in cases of hydrocephalus there will be damages the hypothalamus making the GnRH-secreting neurones functionless, thereby creating a hypogonadotrophic hypogonadic state.

Trauma to the skull base may also damage the hypothalamus and prevent GnRH secretion.

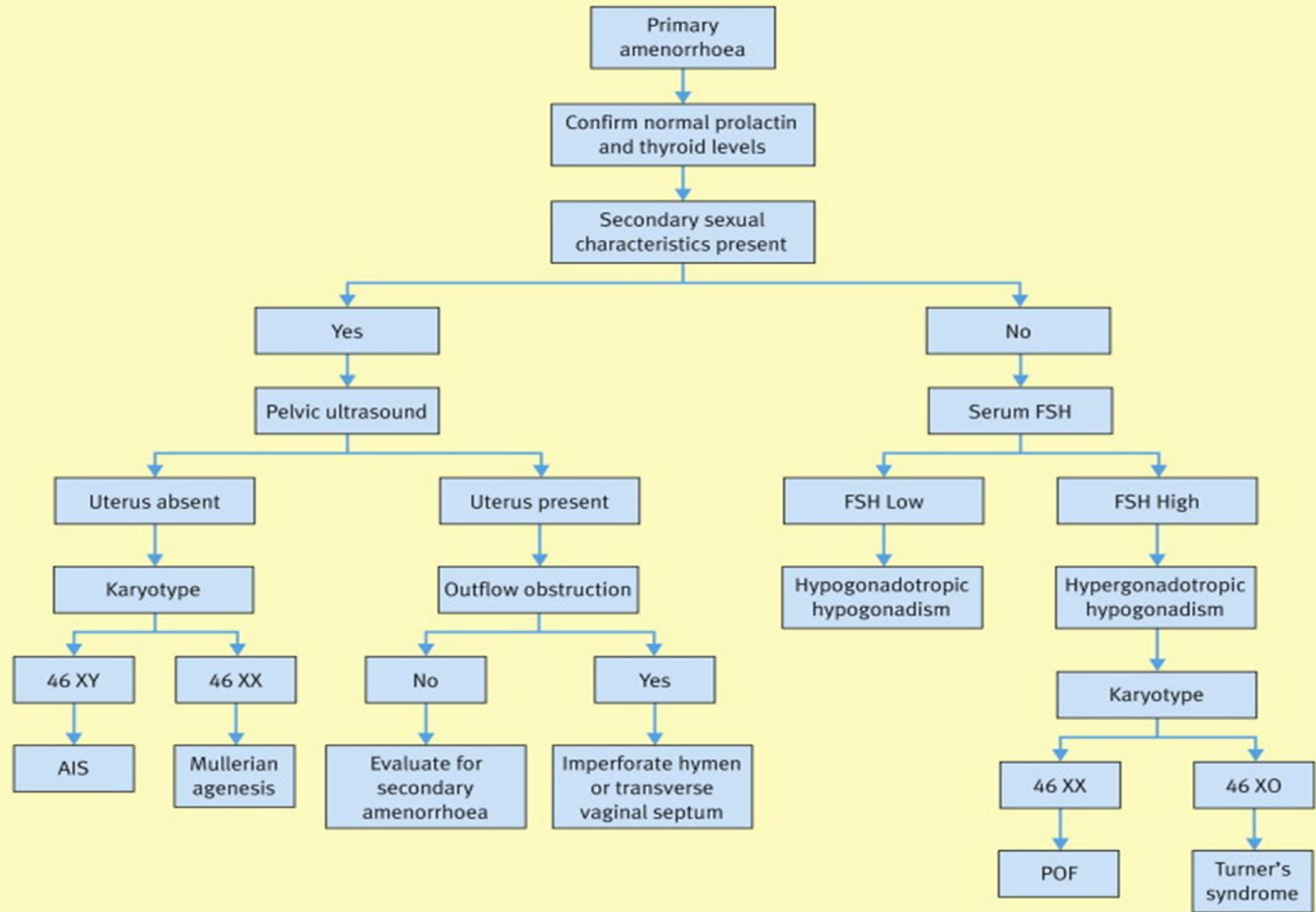
Empty sella syndrome

- the sella turcica is empty and there is congenital absence of the pituitary gland or at least part of it, leading to failure to produce gonadotrophins.

Tumours

- craniopharyngioma results in destruction of the pituitary gland.

Algorithm for investigation of primary amenorrhoea



Management of primary amenorrhea

- The aim of treatment is to attain maximum physiologic function of which an individual can attain.
- No treatment until the diagnosis is solid.
- The majority will be able to achieve satisfactory sexual life but the prospect of fertility may be poor.
- HRT may be needed for life.

- Turner's Syndrome (growth hormone first –estrogen replacement later).
- Rokitansky Syndrome –vaginoplasty
- XY female(remove gonads –Estrogen replacement –Vaginoplasty)
- Imperforated hymen need hymenotomy.
- Vaginal outflow obstruction needs vaginoplasty.

Causes of secondary amenorrhoea:

- **Uterine causes** -Asherman's syndrome, Cervical stenosis.
- **Ovarian causes** Polycystic ovary syndrome, Premature ovarian failure (genetic, autoimmune, infective, radio/ chemotherapy.
- **Hypothalamic causes (hypogonadotrophic hypogonadism):** Weight loss, Exercise, Chronic illness, Psychological distress, Idiopathic.
- **Pituitary causes :**Hyperprolactinaemia, Hypopituitarism, Sheehan's syndrome.
- **Causes of hypothalamic/ pituitary damage, (hypogonadism):** Tumours (craniopharyngiomas, gliomas, germinomas, dermoid cysts), Cranial irradiation, Head injuries, Sarcoidosis, Tuberculosis, Chronic debilitating illness.
- **Systemic causes :**Weight loss, Endocrine disorders (thyroid disease, Cushing's syndrome, etc.)

Assessment for patients with secondary amenorrhoea:

- A thorough history and a careful examination looking particularly at stature and body form, signs of endocrine disease, secondary sexual development and the external genitalia.
- A history of secondary amenorrhoea may be misleading, as in (HRT) for primary amenorrhoea.
- A family history of fertility problems, autoimmune disorders or premature menopause may also give clues as to the aetiology.

- Is there is any sings of hyperandrogenism (hirsutism ,acne and scalp hair loss) and virilization, (high circulating androgen levels and causes deepening of the voice, breast atrophy, increase in muscle bulk and cliteromegaly).
- A rapid onset of hirsutism suggests the possibility of an androgen-secreting tumour of the ovary or adrenal gland.
- Cushing's syndrome
- Acanthosis nigricans is a sign of profound insulin resistance

- Prescence of galactorrhea may indicate hyperprolactinemia, bilateral hemianopia that suggest pituitary gland tumor compressing the optic chiasma so brain MRI is indicated.
- Thyroid disease is common and the thyroid gland should be palpated and signs of hypothyroidism (dry thin hair, proximal myopathy, myotonia, slow-relaxing reflexes, mental slowness, bradycardia, etc.) or hyperthyroidism (goitre with bruit, tremor, weight loss, tachycardia, hyper-reflexia, exophthalmos, conjunctival oedema, ophthalmoplegia) elicited.

- The patient's oestrogen status may be assessed clinically by examination of the lower genital tract or by means of a progestogen challenge.
- Serum measurements of oestradiol are of limited value
- If the patient is well oestrogenized, the endometrium will be clearly seen on an ultrasound scan and should be shed on withdrawal of the progestogen.
- Recent history of endometrial curettage with subsequent amenorrhea is suggestive for asherman syndrome.
- Serum gonadotrophin measurements help to distinguish between cases of hypothalamic or pituitary failure and gonadal failure.

Investigation of secondary amenorrhea:

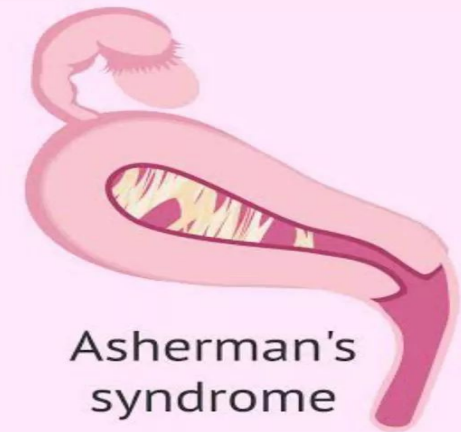
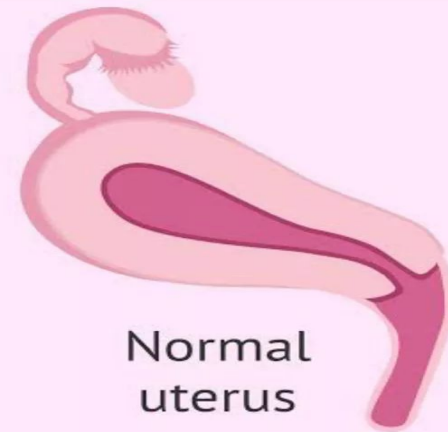
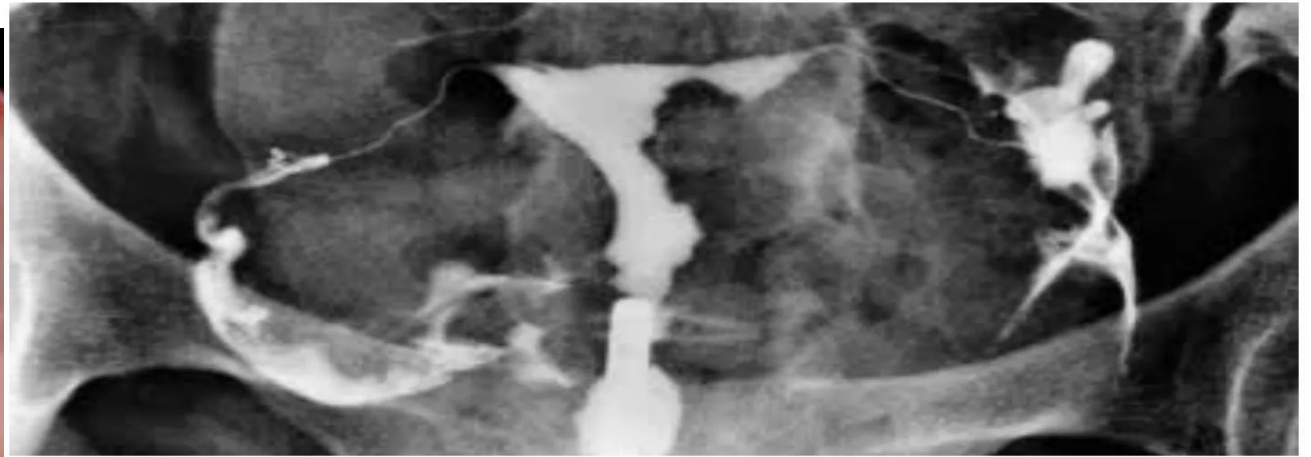
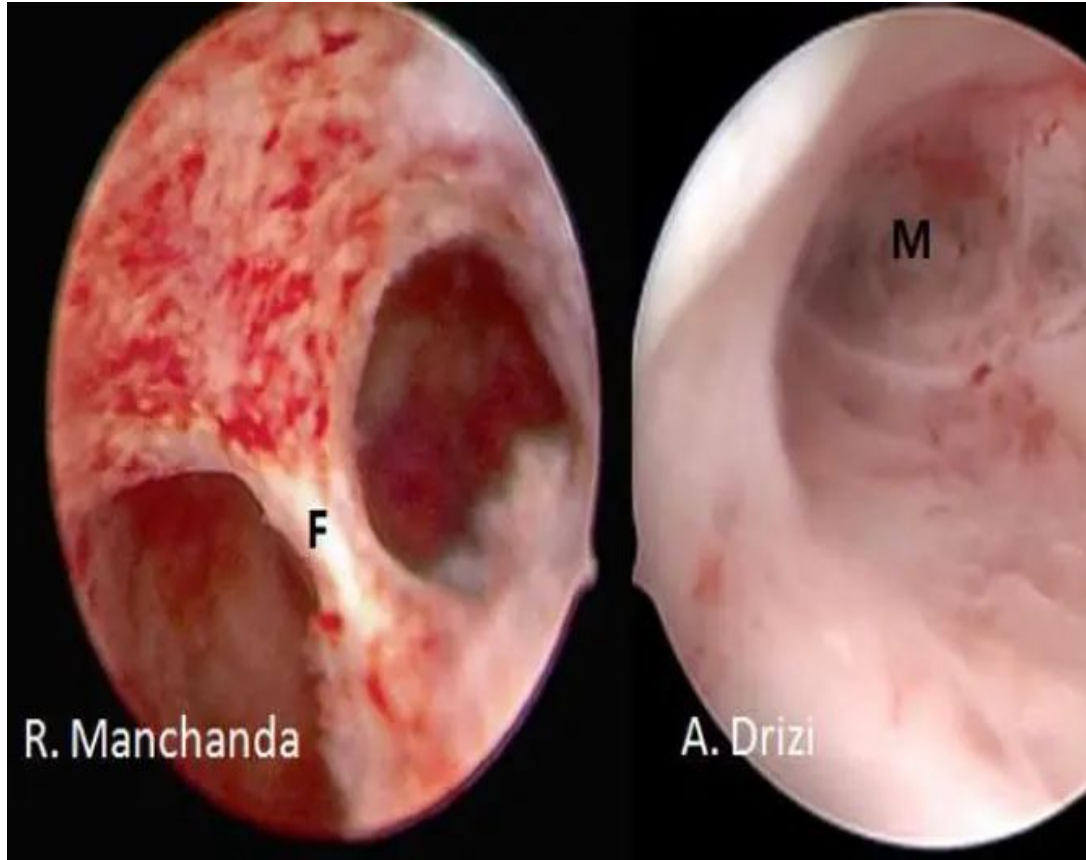
- A pregnancy test should be carried out if the patient is sexually active.
- LH, FSH (early follicular phase of cycle) and testosterone; raised LH or raised testosterone could be suggestive of PCOS; raised FSH may be suggestive of POF.
- A raised prolactin level may indicate a prolactinoma.
- Thyroid function should be checked if clinically indicated.
- An ultrasound scan (polycystic ovaries) and (MRI) of the brain should be carried out if symptoms are consistent with a pituitary adenoma.
- Hysteroscopy is not routine, but is a suitable investigation where Asherman or cervical stenosis is suspected.

Management of secondary amenorrhoea

Asherman's syndrome

- This is a result of an over-vigorous endometrial curettage affecting the basalis layer of the endometrium growth or adhesions that may follow an episode of endometritis.
- Typically, amenorrhoea is not absolute, and it may be possible to induce a withdrawal bleed using a combined oestrogen/progestogen preparation.
- Intrauterine adhesions may be seen on an HSG .
- Alternatively, hysteroscopic inspection of the uterine cavity will confirm the diagnosis and enable treatment by adhesiolysis.

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- Following surgery, a 3-month course of cyclical progesterone/oestrogen should be given.
- Some clinicians insert a foley catheter into the uterine cavity for 7–10 days postoperatively or an intrauterine contraceptive device for 2–3 months in order to prevent recurrence of adhesions.



Cervical stenosis: It was relatively common following a traditional cone biopsy for the treatment of cervical intraepithelial neoplasia. Treatment for cervical stenosis consists of careful cervical dilatation.

Pituitary causes of secondary amenorrhoea

- Hyperprolactinaemia is the commonest pituitary cause of amenorrhoea.
- Other causes include hypothyroidism, PCOS (up to 2500 mIU/L) and several drugs (e.g. the dopaminergic antagonist phenothiazines, domperidone and metoclopramide).
- In women with amenorrhoea associated with **hyperprolactinaemia**, the main symptoms are usually those of oestrogen deficiency.

- **The management of hyperprolactinaemia** is dopamine agonist, of which bromocriptine is the most widely used. Longer acting preparations (e.g. twice-weekly cabergoline)
- may be prescribed to those patients who develop unacceptable side effects.
- Surgery, in the form of a trans-sphenoidal adenectomy (macroadenoma or if there are intolerable side effects of the drugs).
- If the patient has a regular menstrual cycle, no treatment is necessary unless the cycle is anovulatory and fertility is desired.
- Amenorrhoea is the 'bioassay' of prolactin excess and should be corrected for its sequelae, rather than for the serum level of prolactin.

Sheehan's syndrome, the result of profound and prolonged hypotension on the sensitive pituitary gland, enlarged by pregnancy, as in cases of major obstetric haemorrhage.

- It is essential to assess the pituitary function fully in all these patients and then instigate the appropriate replacement therapy.
- Ovulation may be induced with pulsatile subcutaneous GnRH or human menopausal gonadotrophins (hMG).

Premature ovarian failure: Management will be by using HRT and change life style .

Management of PCOS involves the following:

- Combined oral contraceptive pill (COCP) to regulate menstruation. This also increases sex hormone binding globulin, which will help reduce androgenic symptoms.
- Cyclical oral progesterone: used to regulate a withdrawal bleed.
- Clomiphene: this can be used to induce ovulation where subfertility is a factor.
- Lifestyle advice: dietary modification and exercise is appropriate in these patients as they are at an increased risk of developing diabetes and cardiovascular disease later in life. Aerobic exercise has been shown to improve insulin resistance.

Hypothalamic causes of secondary amenorrhoea:

- may be either primary or secondary.
- Primary hypothalamic lesions include craniopharyngiomas, germinomas, gliomas and dermoid cysts. These hypothalamic lesions either disrupt the normal pathway of prolactin inhibitory factor (dopamine), thus causing hyperprolactinaemia, or compress and/or destroy hypothalamic and pituitary tissue.
- Treatment is usually **surgical, with additional radiotherapy if required.**
- HRT is required to mimic ovarian function, and if the pituitary gland is damaged either by the lesion or by the treatment, replacement thyroid and adrenal hormones are required.

- Secondary hypogonadotrophic hypogonadism (HH) may result from systemic conditions including sarcoidosis and tuberculosis as well as following head injury or cranial irradiation.

Weight-related amenorrhoea:

- Weight can have profound effects on gonadotrophin regulation and release.
- BMI is $<19 \text{ kg/m}^2$.
- 22% of body weight should be fat in order to maintain ovulatory cycles because this enables the extraovarian aromatization of androgens to oestrogens and maintains appropriate feedback control of the hypothalamic–pituitary–ovarian (HPO) axis.
- An artificial cycle may be induced with the combined oral contraceptive pill and wt mx .

Systemic disorders causing secondary amenorrhoea

- Chronic disease may result in menstrual disorders as a consequence of the general disease state, weight loss or by the effect of the disease process on the hypothalamic– pituitary axis.
- increase the risk of amenorrhoea-associated osteoporosis(COPD)
- Women with chronic renal failure have a discordantly elevated LH, possibly as a consequence of impaired clearance.
- Liver disease affects the level of circulating sex hormone-binding globulin and thus hormone levels, thereby disrupting the normal feedback mechanisms.

- Metabolism of various hormones including testosterone are also liver-dependent
- Endocrine disorders such as thyrotoxicosis and Cushing's syndrome are commonly associated with gonadal dysfunction.
- Autoimmune endocrinopathies may be associated with POF because of ovarian antibodies.
- Diabetes mellitus may result in functional hypothalamic–pituitary amenorrhoea.
- Management of these patients should concentrate on the underlying systemic problem and on preventing complications of oestrogen deficiency

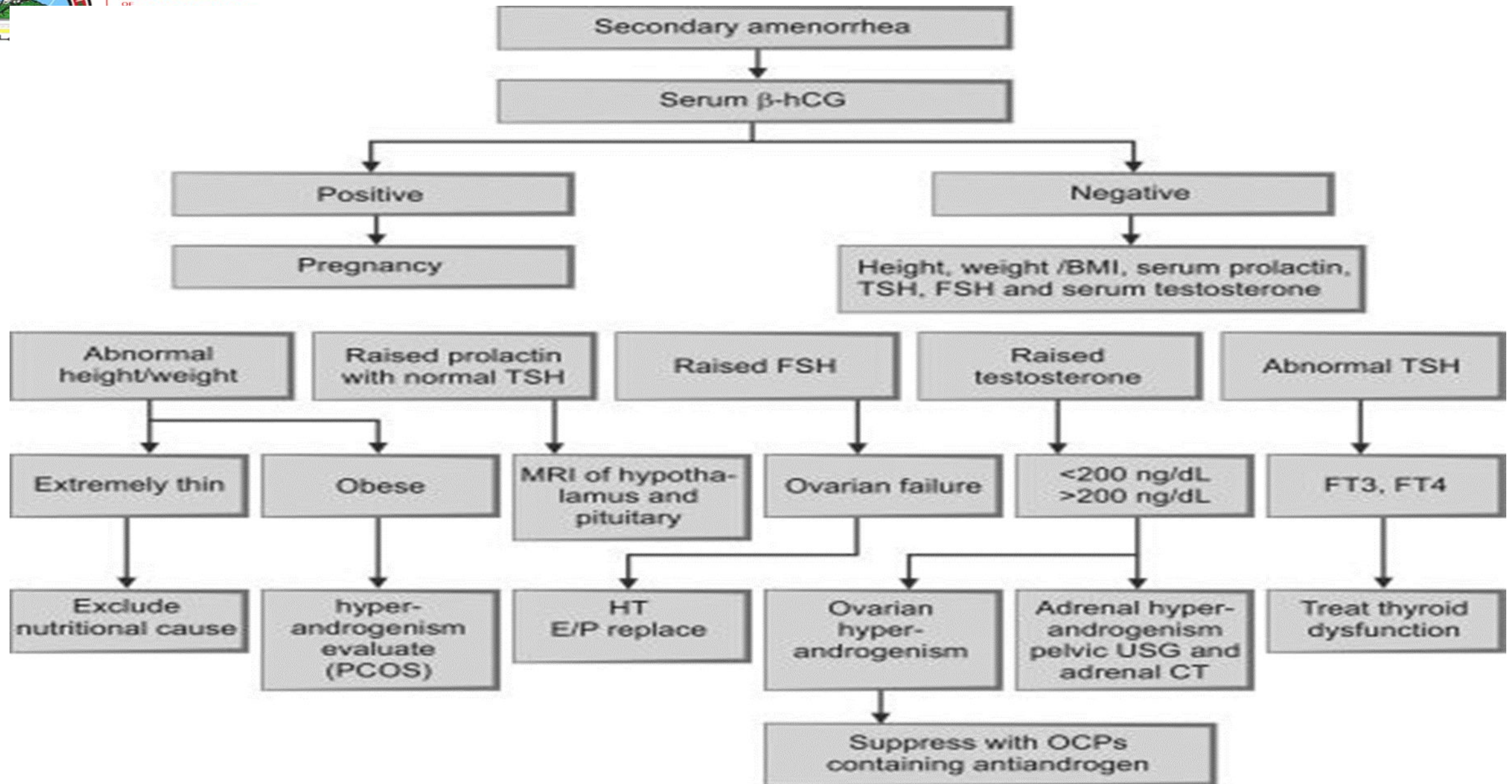


Table 3.3 History and examination of patients with amenorrhoea/oligomenorrhoea

Information required	Relevant factors	Possible diagnoses
Developmental history including menarche	Delayed/incomplete puberty	Congenital malformation or chromosomal abnormality
Menstrual history	Oligomenorrhoea	PCOS
	Secondary amenorrhoea	POF
Reproductive history	Infertility	PCOS Congenital malformation
Cyclical symptoms	Cyclical pain without menstruation	Congenital malformation
		Imperforate hymen
Hair growth	Hirsutism, androgenic alopecia	PCOS
	Hair loss	Hypothyroidism
Weight	Dramatic weight loss	Hypothalamic malfunction, hormone producing malignancy
	Difficulty losing weight	PCOS, hypothyroidism
Lifestyle	Exercise, stress	Hypothalamic malfunction
Past medical history	Systemic diseases (e.g. sarcoidosis)	Hypothalamic malfunction
Past surgical history	Evacuation of uterus	Asherman syndrome, cervical stenosis
Drug history	Dopamine agonists, HRT	Hypothalamic malfunction
	Past chemotherapy	POF
Headache, visual disturbance		Pituitary adenoma
Galactorrhoea		Prolactinoma

HRT, hormone replacement therapy; PCOS, polycystic ovary syndrome; POF, premature ovarian failure.

Cause of amenorrhoea/oligomenorrhoea	Management
Low BMI	Dietary advice and support
Hypothalamic lesions, e.g. glioma	Surgery
Hyperprolactinaemia/prolactinoma	Dopamine agonist (e.g. cabergoline or bromocriptine) or surgery if medication fails
POF	HRT or COCP
PCOS	See below
Asherman's	Adhesiolysis and IUD insertion at time of hysteroscopy (to prevent recurrence of adhesions)
Cervical stenosis	Hysteroscopy and cervical dilatation



Thank you



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